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SIGNAL PROCESSING DEVICE AND VEHICLE DISPLAY DEVICE COMPRISING SAME

Abstract

The signal processing device and the vehicle display apparatus including the same according to an embodiment of the present disclosure includes a processor to process a vehicle signal, wherein the processor is configured to execute a server virtual machine and a plurality of guest virtual machines on a hypervisor in the processor, wherein the server virtual machine is configured to transmit processed camera data to a shared memory, a second guest virtual machine is configured to receive the camera data from the shared memory, to generate first data for a first application service based on the camera data and to transmit the first data to the shared memory, and a first guest virtual machine is configured to display an image, synthesized based on the camera data from the shared memory and the first data, on the first display. Accordingly, data processing may be performed efficiently.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/721,153,filed on Jun. 17, 2024, which is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2022/016126, filed on Oct. 21, 2022, the contents of which are all hereby incorporated by reference herein their entirety.

BACKGROUND
1. Technical Field
[0002] The present disclosure relates to a signal processing device and a vehicle display apparatus including the same, and more particularly to a signal processing device capable of efficiently performing data processing, and a vehicle display apparatus including the signal processing device.
2. Description of the Related Art
[0003] A vehicle is a device that allows a user to move in desired directions. A typical example of the vehicle is a car.
[0004] Meanwhile, a vehicle display apparatus is mounted in the vehicle for user convenience.
[0005] For example, a display mounted on a cluster and the like displays a variety of information. Meanwhile, in order to display vehicle driving information and the like, various displays, such as an Audio Video Navigation (AVN) display, etc., are mounted in the vehicle separately from the cluster.
[0006] However, as the number of displays in the vehicle display apparatus increases, a problem occurs in that signal processing for the displays becomes complicated.
SUMMARY
[0007] It is an objective of the present disclosure to provide a signal processing device capable of efficiently performing data processing, and a vehicle display apparatus including the signal processing device.
[0008] Meanwhile, it is another objective of the present disclosure to provide a signal processing device capable of efficiently performing image data processing based on camera data, and a vehicle display apparatus including the signal processing device.
[0009] Meanwhile, it is yet another objective of the present disclosure to provide a signal processing device capable of efficiently performing data processing even when a plurality of virtual machines run on different operating systems, and a vehicle display apparatus including the signal processing device.
[0010] In accordance with an aspect of the present disclosure, the above and other objectives can be accomplished by providing a signal processing device including a processor configured to perform signal processing for a plurality of displays located in a vehicle, wherein the processor is configured to execute a server virtual machine and a plurality of guest virtual machines on a hypervisor in the processor, wherein a first guest virtual machine and a second guest virtual machine among the plurality of guest virtual machines are configured to operate for a first display and a second display, respectively, wherein the server virtual machine is configured to process input camera data and transmit the processed camera data to a shared memory set based on the hypervisor, wherein the second guest virtual machine is configured to receive the camera data from

the shared memory, to generate first data for a first application service based on the camera data, and to transmit the first data to the shared memory; and the first guest virtual machine is configured to display an image, synthesized based on the camera data from the shared memory and the first data, on the first display.

[0011] Meanwhile, the first guest virtual machine may be configured to operate for the first display based on a first operating system, and the second guest virtual machine may be configured to operate for the second display based on a second operating system different from the first operating system.

[0012] Meanwhile, the second guest virtual machine may be configured to display a second image, synthesized based on the camera data from the shared memory and the first data, on the second display.

[0013] Meanwhile, the first guest virtual machine may be configured to transmit the synthesized image to the shared memory; and the second guest virtual machine may be configured to display the synthesized image from the shared memory on the second display.

[0014] Meanwhile, a third guest virtual machine among the plurality of guest virtual machines may be configured to receive the camera data from the shared memory, to generate second data for a second application service based on the camera data, and to transmit the second data to the shared memory, wherein the first guest virtual machine may be configured to display a third image, synthesized based on the camera data from the shared memory and the first data and the second data, on the first display.

[0015] Meanwhile, the first guest virtual machine may be configured to operate for the first display based on a first operating system, and the second guest virtual machine may be configured to operate for the second display based on a second operating system different from the first operating system, and the third guest virtual machine may be configured to operate for the third display based on a third operating system different from the first operating system and the second operating system.

[0016] Meanwhile, the second guest virtual machine may be configured to display a second image, synthesized based on the camera data from the shared memory and the first data, on the second display; and the third guest virtual machine may be configured to display a fourth image, synthesized based on the camera data from the shared memory and the second data, on the third display.

[0017] Meanwhile, the second guest virtual machine may be configured to display a fifth image, synthesized based on the camera data from the shared memory and the first data and the second data, on the second display; and the third guest virtual machine may be configured to display a sixth image, synthesized based on the camera data from the shared memory and the first data and the second data, on the third display.

[0018] Meanwhile, the first guest virtual machine may be configured to transmit the synthesized third image to the shared memory, wherein at least one of the second guest virtual machine or the third guest virtual machine may be configured to display the synthesized third image from the shared memory on at least one of the second display or the third display.

[0019] Meanwhile, an application manager in the first guest virtual machine or the second guest virtual machine may be configured to transmit an event request to the server virtual machine, wherein the server virtual machine may be configured to transmit sharable camera data among the processed camera data to the shared memory based on the event request.

[0020] Meanwhile, the server virtual machine may be configured to execute a data sharing manager for sharing the camera data and a camera processing manager for processing the camera data.

[0021] In accordance with another aspect of the present disclosure, there is provided a signal processing device including a processor configured to perform signal processing for a plurality of displays located in a vehicle, wherein the processor is configured to execute a server virtual machine and first and second guest virtual machines that operate on different operating systems on a hypervisor in the processor, wherein a first guest virtual machine and a second guest virtual machine are configured to operate for a first display and a second display, respectively, wherein the server virtual machine is configured to process input data and transmit the processed shared data to a shared memory set based on the hypervisor, wherein the second guest virtual machine is configured to receive the shared data from the shared memory, to generate first data for a first application service, and to transmit the first data to the shared memory; and the first guest virtual machine is configured to output content, synthesized based on the shared data from the shared memory and the first data, to the first display or an audio output device.

[0022] Meanwhile, the shared data may include at least one of camera data, radar data, lidar data, audio data, or image data.

[0023] Meanwhile, the synthesized content may include at least one of a synthesized image or a synthesized sound.

[0024] In accordance with yet another aspect of the present disclosure, there is provided a vehicle display apparatus including: a plurality of displays; and a signal processing device including a processor configured to perform signal processing for the plurality of displays, wherein the processor is configured to execute a server virtual machine and a plurality of guest virtual machines on a hypervisor in the processor, wherein a first guest virtual machine and a second guest virtual machine among the plurality of guest virtual machines are configured to operate for a first display and a second display, respectively, wherein the server virtual machine is configured to process input camera data and transmit the processed camera data to a shared memory set based on the hypervisor, wherein the second guest virtual machine is configured to receive the camera data from the shared memory, to generate first data for a first application service based on the camera data, and to transmit the first data to the shared memory; and the first guest virtual machine is configured to display an image, synthesized based on the camera data from the shared memory and the first data, on the first display.

EFFECTS OF THE DISCLOSURE

[0025] A signal processing device according to an embodiment of the present disclosure includes a processor configured to perform signal processing for a plurality of displays located in a vehicle, wherein the processor is configured to execute a server virtual machine and a plurality of guest virtual machines on a hypervisor in the processor, wherein a first guest virtual machine and a second guest virtual machine among the plurality of guest virtual machines are configured to operate for a first display and a second display, respectively, wherein the server virtual machine is configured to process input camera data and transmit the processed camera data to a shared memory set based on the hypervisor, wherein the second guest virtual machine is configured to receive the camera data from the shared memory, to generate first data for a first application service based on the camera data, and to transmit the first data to the shared memory; and the first guest virtual machine is configured to display an image, synthesized based on the camera data from the shared memory and the first data, on the first display. Accordingly, data processing may be performed efficiently. Particularly, image data processing based on camera data may be performed efficiently.

[0026] Meanwhile, the first guest virtual machine may be configured to operate for the first display based on a first operating system, and the second guest virtual machine may be configured to operate for the second display based on a second operating system different from the first operating system. Accordingly, even when the plurality of virtual machines run on different operating systems, image data processing based on camera data may be performed efficiently.

[0027] Meanwhile, the second guest virtual machine may be configured to display a second image, synthesized based on the camera data from the shared memory and the first data, on the second display. Accordingly, image data processing based on camera data may be performed efficiently.

[0028] Meanwhile, the first guest virtual machine may be configured to transmit the synthesized image to the shared memory; and the second guest virtual machine may be configured to display

the synthesized image from the shared memory on the second display. Accordingly, image data processing based on camera data may be performed efficiently.
[0029] Meanwhile, a third guest virtual machine among the plurality of guest virtual machines may be configured to receive the camera data from the shared memory, to generate second data for a second application service based on the camera data, and to transmit the second data to the shared memory, wherein the first guest virtual machine may be configured to display a third image, synthesized based on the camera data from the shared memory and the first data and the second data, on the first display. Accordingly, image data processing based on camera data may be performed efficiently.
[0030] Meanwhile, the first guest virtual machine may be configured to operate for the first display based on a first operating system, and the second guest virtual machine may be configured to operate for the second display based on a second operating system different from the first operating system, and the third guest virtual machine may be configured to operate for the third display based on a third operating system different from the first operating system and the second operating system. Accordingly, even when the plurality of virtual machines run on different operating systems, image data processing based on camera data may be performed efficiently.
[0031] Meanwhile, the second guest virtual machine may be configured to display a second image, synthesized based on the camera data from the shared memory and the first data, on the second display; and the third guest virtual machine may be configured to display a fourth image, synthesized based on the camera data from the shared memory and the second data, on the third display. Accordingly, image data processing based on camera data may be performed efficiently.
[0032] Meanwhile, the second guest virtual machine may be configured to display a fifth image, synthesized based on the camera data from the shared memory and the first data and the second data, on the second display; and the third guest virtual machine may be configured to display a sixth image, synthesized based on the camera data from the shared memory and the first data and the second data, on the third display. Accordingly, image data processing based on camera data may be performed efficiently.
[0033] Meanwhile, the first guest virtual machine may be configured to transmit the synthesized third image to the shared memory, wherein at least one of the second guest virtual machine or the third guest virtual machine may be configured to display the synthesized third image from the shared memory on at least one of the second display or the third display. Accordingly, image data processing based on camera data may be performed efficiently.
[0034] Meanwhile, an application manager in the first guest virtual machine or the second guest virtual machine may be configured to transmit an event request to the server virtual machine, wherein the server virtual machine may be configured to transmit sharable camera data among the processed camera data to the shared memory based on the event request. Accordingly, image data processing based on camera data may be performed efficiently.
[0035] Meanwhile, the server virtual machine may be configured to execute a data sharing manager for sharing the camera data and a camera processing manager for processing the camera data. Accordingly, image data processing based on camera data may be performed efficiently.
[0036] In accordance with another aspect of the present disclosure, there is provided a signal processing device including a processor configured to perform signal processing for a plurality of displays located in a vehicle, wherein the processor is configured to execute a server virtual machine and first and second guest virtual machines that operate on different operating systems on a hypervisor in the processor, wherein a first guest virtual machine and a second guest virtual machine are configured to operate for a first display and a second display, respectively, wherein the server virtual machine is configured to process input data and transmit the processed shared data to a shared memory set based on the hypervisor, wherein the second guest virtual machine is configured to receive the shared data from the shared memory, to generate first data for a first application service, and to transmit the first data to the shared memory; and the first guest virtual

machine is configured to output content, synthesized based on the shared data from the shared memory and the first data, to the first display or an audio output device. Accordingly, data processing may be performed efficiently.
[0037] Meanwhile, the shared data may include at least one of camera data, radar data, lidar data, audio data, or image data. Accordingly, data processing may be performed efficiently.
[0038] Meanwhile, the synthesized content may include at least one of a synthesized image or a synthesized sound. Accordingly, data processing may be performed efficiently.
[0039] In accordance with yet another aspect of the present disclosure, there is provided a vehicle display apparatus including: a plurality of displays; and a signal processing device including a processor configured to perform signal processing for the plurality of displays, wherein the processor is configured to execute a server virtual machine and a plurality of guest virtual machines on a hypervisor in the processor, wherein a first guest virtual machine and a second guest virtual machine among the plurality of guest virtual machines are configured to operate for a first display and a second display, respectively, wherein the server virtual machine is configured to process input camera data and transmit the processed camera data to a shared memory set based on the hypervisor, wherein the second guest virtual machine is configured to receive the camera data from the shared memory, to generate first data for a first application service based on the camera data, and to transmit the first data to the shared memory; and the first guest virtual machine is configured to display an image, synthesized based on the camera data from the shared memory and the first data, on the first display. Accordingly, data processing may be performed efficiently. Particularly, image data processing based on camera data may be performed efficiently.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0040] FIG. **1**A is a view showing an example of the exterior and interior of a vehicle;
[0041] FIG. **1**B is a view showing another example of the interior of the vehicle;
[0042] FIG. **2** is a view showing the external appearance of a display apparatus for vehicles according to an embodiment of the present disclosure;
[0043] FIG. **3** illustrates an example of an internal block diagram of the display apparatus for vehicles of FIG. **2**;
[0044] FIG. **4** is a view showing a system executed in a signal processing device according to an embodiment of the present disclosure;
[0045] FIG. **5** is a diagram showing an example of a system executed in a signal processing device according to the present disclosure;
[0046] FIG. **6** is a diagram illustrating another example of a system executed in a signal processing device according to an embodiment of the present disclosure;
[0047] FIG. **7** is a diagram illustrating a further example of a system executed in a signal processing device according to an embodiment of the present disclosure;
[0048] FIGS. **8** to **9**B are diagrams referred to in the description of FIG. **5**;
[0049] FIG. **10** is an exemplary internal block diagram of a vehicle display apparatus according to an embodiment of the present disclosure; and
[0050] FIGS. **11** to **15**C are diagrams referred to in the description of FIG. **10**.
DETAILED DESCRIPTION
[0051] Hereinafter, the present disclosure will be described in detail with reference to the accompanying drawings.
[0052] With respect to constituent elements used in the following description, suffixes “module” and “unit” are given only in consideration of ease in preparation of the specification, and do not have or serve different meanings. Accordingly, the suffixes “module” and “unit” may be used

interchangeably.
[0053] FIG. **1**A is a view showing an example of the exterior and interior of a vehicle.
[0054] Referring to the figure, the vehicle **200** is moved by a plurality of wheels **103**FR, **103**FL, **103**RL, . . . rotated by a power source and a steering wheel **150** configured to adjust an advancing direction of the vehicle **200**.
[0055] Meanwhile, the vehicle **200** may be provided with a camera **195** configured to acquire an image of the front of the vehicle.
[0056] Meanwhile, the vehicle **200** may be further provided therein with a plurality of displays **180***a* and **180***b* configured to display images and information.
[0057] In FIG. **1**A, a cluster display **180***a* and an audio video navigation (AVN) display **180***b* are illustrated as the plurality of displays **180***a* and **180***b*. In addition, a head up display (HUD) may also be used.
[0058] Meanwhile, the audio video navigation (AVN) display **180***b* may also be called a center information display.
[0059] The embodiment of the present disclosure proposes a scheme for dividing data processing in a vehicle display apparatus **100** including a plurality of displays **180***a* and **180***b*, which will be described later with reference to FIG. **12** and subsequent figures.
[0060] Meanwhile, the vehicle **200** described in this specification may be a concept including all of a vehicle having an engine as a power source, a hybrid vehicle having an engine and an electric motor as a power source, and an electric vehicle having an electric motor as a power source.
[0061] FIG. **1**B is a view showing another example of the interior of the vehicle.
[0062] Referring to the figure, a cluster display **180***a*, an audio video navigation (AVN) display **180***b*, rear seat entertainment displays **180***c* and **180***d*, and a rear-view mirror display (not shown) may be located in the vehicle **200**.
[0063] The embodiment of the present disclosure proposes a scheme for efficiently performing data processing in a vehicle display apparatus **100** including a plurality of displays **180***a* and **180***b*, which will be described later with reference to FIG. **5** and subsequent figures.
[0064] FIG. **2** is a view showing the external appearance of a display apparatus for vehicles according to an embodiment of the present disclosure.
[0065] The display apparatus **100** for vehicles according to the embodiment of the present disclosure may include a plurality of displays **180***a* and **180***b* and a signal processing device **170** configured to perform signal processing in order to display images and information on the plurality of displays **180***a* and **180***b*.
[0066] The first display **180***a*, which is one of the plurality of displays **180***a* and **180***b*, may be a cluster display **180***a* configured to display a driving state and operation information, and the second display **180***b* may be an audio video navigation (AVN) display **180***b* configured to display vehicle driving information, a navigation map, various kinds of entertainment information, or an image.
[0067] The signal processing device **170** may have a processor **175** provided therein, and first to third virtual machines **520** to **540** may be executed by a hypervisor **505** in the processor **175**.
[0068] The second virtual machine **530** may be operated for the first display **180***a*, and the third virtual machine **540** may be operated for the second display **180***b*.
[0069] Meanwhile, the first virtual machine **520** in the processor **175** may perform control such that a shared memory **508** based on the hypervisor **505** is set for transmission of the same data to the second virtual machine **530** and the third virtual machine **540**. Consequently, the first display **180***a* and the second display **180***b* in the vehicle may display the same information or the same images in a synchronized state.
[0070] Meanwhile, the first virtual machine **520** in the processor **175** shares at least some of data with the second virtual machine **530** and the third virtual machine **540** for divided processing of data. Consequently, the plurality of virtual machines for the plurality of displays in the vehicle may divide and process data.

[0071] Meanwhile, the first virtual machine **520** in the processor **175** may receive and process wheel speed sensor data of the vehicle, and may transmit the processed wheel speed sensor data to at least one of the second virtual machine **530** or the third virtual machine **540**. Consequently, at least one virtual machine may share the wheel speed sensor data of the vehicle.
[0072] Meanwhile, the display apparatus **100** for vehicles according to the embodiment of the present disclosure may further include a rear seat entertainment (RSE) display **180***c* configured to display driving state information, simple navigation information, various kinds of entertainment information, or an image.
[0073] The signal processing device **170** may further execute a fourth virtual machine (not shown), in addition to the first to third virtual machines **520** to **540**, on the hypervisor **505** in the processor **175** to control the RSE display **180***c*.
[0074] Consequently, it is possible to control various displays **180***a* to **180***c* using a single signal processing device **170**.
[0075] Meanwhile, some of the plurality of displays **180***a* to **180***c* may be operated based on a Linux Operating System (OS), and others may be operated based on a Web Operating System (OS).
[0076] When touch is input to any one of the displays **180***a* and **180***b* or **180***a* to **180***c* configured to be operated under various operating systems, the signal processing device **170** according to the embodiment of the present disclosure may perform control such that the touch input is rapidly and accurately processed.
[0077] Meanwhile, FIG. **2** illustrates that a vehicle speed indicator **212***a* and an in-vehicle temperature indicator **213***a* are displayed on the first display **180***a*, a home screen **222** including a plurality of applications, a vehicle speed indicator **212***b*, and an in-vehicle temperature indicator **213***b* is displayed on the second display **180***b*, and a home screen **222***b* including a plurality of applications and an in-vehicle temperature indicator **213***c* is displayed on the third display **180***c*.
[0078] FIG. **3** illustrates an example of an internal block diagram of the display apparatus for vehicles according to the embodiment of the present disclosure.
[0079] Referring to the figure, the display apparatus **100** for vehicles according to the embodiment of the present disclosure may include an input device **110**, a transceiver **120**, an interface **130**, a memory **140**, a signal processing device **170**, a plurality of displays **180***a* to **180***c*, an audio output device **185**, and a power supply **190**.
[0080] The input device **110** may include a physical button or pad for button input or touch input.
[0081] Meanwhile, the input device **110** may include a touch sensor (not shown) configured to sense touch input to the displays **180***a*, **180***b*, and **180***c*.
[0082] Meanwhile, the input device **110** may include a microphone (not shown) for user voice input.
[0083] The transceiver **120** may wirelessly exchange data with a mobile terminal **800** or a server **900**.
[0084] In particular, the transceiver **120** may wirelessly exchange data with a mobile terminal **800** of a vehicle driver. Any of various data communication schemes, such as Bluetooth, Wi-Fi, WIFI Direct, and APIX, may be used as a wireless data communication scheme.
[0085] The transceiver **120** may receive weather information and road traffic situation information, such as transport protocol expert group (TPEG) information, from the mobile terminal **800** or the server **900**. To this end, the transceiver **120** may include a mobile communication module (not shown).
[0086] The interface **130** may receive sensor information from an electronic control unit (ECU) **770** or a sensor device **760**, and may transmit the received information to the signal processing device **170**.
[0087] Here, the sensor information may include at least one of vehicle direction information, vehicle position information (global positioning system (GPS) information), vehicle angle information, vehicle velocity information, vehicle acceleration information, vehicle inclination

information, vehicle forward/backward movement information, battery information, fuel information, tire information, vehicle lamp information, in-vehicle temperature information, or in-vehicle humidity information.
[0088] The sensor information may be acquired from a heading sensor, a yaw sensor, a gyro sensor, a position sensor, a vehicle forward/backward movement sensor, a wheel sensor, a vehicle velocity sensor, a car body inclination sensor, a battery sensor, a fuel sensor, a tire sensor, a steering-wheel-rotation-based steering sensor, an in-vehicle temperature sensor, or an in-vehicle humidity sensor. Meanwhile, the position module may include a GPS module configured to receive GPS information.
[0089] Meanwhile, the interface **130** may receive front-of-vehicle image data, side-of-vehicle image data, rear-of-vehicle image data, and obstacle-around-vehicle distance information from a camera **195** or lidar (not shown) and may transmit the received information to the signal processing device **170**.
[0090] The memory **140** may store various data for overall operation of the vehicle display apparatus **100**, including programs for processing or control of the signal processing device **170**.
[0091] For example, the memory **140** may store data about the hypervisor and the first to third virtual machines executed in the processor **175**.
[0092] The audio output device **185** may convert an electrical signal from the signal processing device **170** into an audio signal, and may output the audio signal. To this end, the audio output device **185** may include a speaker.
[0093] The power supply **190** may supply power necessary to operate components under control of the signal processing device **170**. In particular, the power supply **190** may receive power from a battery in the vehicle.
[0094] The signal processing device **170** may control overall operation of each device in the display apparatus **100** for vehicles.
[0095] For example, the signal processing device **170** may include a processor **175** for performing signal processing for the vehicle displays **180***a* and **180***b*.
[0096] The processor **175** may execute the first to third virtual machines **520** to **540** on the hypervisor **505** (see FIG. **5**) in the processor **175**.
[0097] Meanwhile, the processor **175** may further execute a legacy virtual machine configured to receive and process Ethernet data. For example, as shown in FIG. **5**, the legacy virtual machine **510** may be executed by the first virtual machine **520** in the processor **175**.
[0098] Among the first to third virtual machines **520** to **540** (see FIG. **5**), the first virtual machine **520** may be called a server virtual machine, and the second and third virtual machines **530** and **540** may be called guest virtual machines.
[0099] In this case, the second virtual machine **530** may be operated for the first display **180***a*, and the third virtual machine **540** may be operated for the second display **180***b*.
[0100] For example, the first virtual machine **520** in the processor **175** may receive, process, and output vehicle sensor data, position information data, camera image data, audio data, or touch input data. Data processed only by a legacy virtual machine and data processed by the first virtual machine **520** may be distinguished from each other, whereby data processing may be efficiently performed. In particular, the first virtual machine **520** may process most of the data, whereby 1:N data sharing may be achieved.
[0101] As another example, the first virtual machine **520** may directly receive and process CAN communication data, audio data, radio data, USB data, and wireless communication data for the second and third virtual machines **530** and **540**.
[0102] The first virtual machine **520** may transmit the processed data to the second and third virtual machines **530** and **540**.
[0103] Consequently, only the first virtual machine **520**, among the first to third virtual machines **520** to **540**, may receive communication data and external input data, and may perform signal

processing, whereby load in signal processing by the other virtual machines may be reduced and 1:N data communication may be achieved, and therefore synchronization at the time of data sharing may be achieved.

[0104] Meanwhile, the first virtual machine **520** writes some of data in a first shared memory (not shown) so as to be transmitted to the second virtual machine **530** and writes some other of data in the first shared memory (not shown) so as to be transmitted to the third virtual machine **540**. The second virtual machine **530** and the third virtual machine **540** may be configured to process the received data and write the processed data in a second shared memory (not shown). Accordingly, data processing may be performed efficiently.

[0105] At this time, data may be any one of image data, audio data, navigation data, and voice recognition data.

[0106] Meanwhile, the first virtual machine **520** may process some other of data, and may be configured to write the processed data in the second shared memory (not shown). That is, the first virtual machine **520** may perform data processing in addition to the second virtual machine **530** and the third virtual machine **540**.

[0107] Meanwhile, in response to a fourth virtual machine **550** configured to be operated for the third display **180***c* being executed in the processor **175**, the first virtual machine **520** may write some other of data in the first shared memory (not shown), and the fourth virtual machine **550** may process the received data and may be configured to write the processed data in the second shared memory (not shown).

[0108] Meanwhile, the first virtual machine **520** may generate command queues for distributed processing of data in the second virtual machine **530** and the third virtual machine **540**. Consequently, the plurality of virtual machines may divide and process data.

[0109] Meanwhile, in response to the second virtual machine **530** and the third virtual machine **540** sharing the same data, the first virtual machine **520** in the processor **175** may generate one command queue. Consequently, the same data may be synchronized and shared.

[0110] Meanwhile, the first virtual machine **520** may generate command queues corresponding to the number of virtual machines for distributed processing of data.

[0111] Meanwhile, the first virtual machine **520** may be configured to transmit at least some of data to at least one of the second virtual machine **530** or the third virtual machine **540** for distributed processing of data.

[0112] For example, the first virtual machine **520** may allocate the first shared memory (not shown) for transmitting at least some of data to at least one of the second virtual machine **530** or the third virtual machine **540**, and image data processed by the second virtual machine **530** or the third virtual machine **540** may be written in the second shared memory (not shown)

[0113] Meanwhile, the first virtual machine **520** may be configured to write data in the shared memory **508**, whereby the second virtual machine **530** and the third virtual machine **540** share the same data.

[0114] For example, the first virtual machine **520** may be configured to write radio data or wireless communication data in the shared memory **508**, whereby the second virtual machine **530** and the third virtual machine **540** share the same data. Consequently, 1:N data sharing may be achieved.

[0115] Eventually, the first virtual machine **520** may process most of the data, whereby 1:N data sharing may be achieved.

[0116] Meanwhile, the first virtual machine **520** in the processor **175** may be configured to set the shared memory **508** based on the hypervisor **505** in order to transmit the same data to the second virtual machine **530** and the third virtual machine **540**.

[0117] That is, the first virtual machine **520** in the processor **175** may transmit the same data to the second virtual machine **530** and the third virtual machine **540** in a synchronized state using the shared memory **508** based on the hypervisor **505**. Consequently, the plurality of displays **180***a* and **180***b* in the vehicle may display the same images in a synchronized state.

[0118] Meanwhile, the signal processing device **170** may process various signals, such as an audio signal, an image signal, and a data signal. To this end, the signal processing device **170** may be implemented in the form of a system on chip (SOC).
[0119] FIG. **4** is a view showing a system executed in a signal processing device according to an embodiment of the present disclosure.
[0120] Referring to the figure, FIG. **4** is a view illustrating that virtual machines are used for the cluster display **180***a* and the AVN display **180***b*.
[0121] The system **400** executed in the signal processing device of FIG. **4** illustrates that a cluster virtual machine **430** and an AVN virtual machine **440** are executed through a hypervisor **405** in the processor **175**.
[0122] Meanwhile, the system **400** executed in the signal processing device of FIG. **4** illustrates that a legacy virtual machine **410** is also executed on the hypervisor **405** in the processor **175**.
[0123] The legacy virtual machine **410** may include an interface **412** for data communication with the memory **140** and an interface **413** for Ethernet communication.
[0124] Meanwhile, the cluster virtual machine **430** may include an interface **431** for CAN communication, an interface **432** for communication with the interface **412** of the legacy virtual machine **410**, and an interface **433** for communication with the interface **413** of the legacy virtual machine **410**.
[0125] Meanwhile, the AVN virtual machine **440** may include an interface **441** for input and output of audio data, radio data, USB data, and wireless communication data, an interface **442** for communication with the interface **412** of the legacy virtual machine **410**, and an interface **443** for communication with the interface **413** of the legacy virtual machine **410**.
[0126] In the system **400**, there is a disadvantage in that CAN communication data are input and output only in the cluster virtual machine **430**, whereby the CAN communication data cannot be utilized in the AVN virtual machine **440**.
[0127] Also, in the system **400** of FIG. **4**, there is a disadvantage in that audio data, radio data, USB data, and wireless communication data are input and output only in the AVN virtual machine **440**, whereby these data cannot be utilized in the cluster virtual machine **430**.
[0128] Meanwhile, there is a disadvantage in that the cluster virtual machine **430** and the AVN virtual machine **440** must include the interfaces **431** and **432** and the interfaces **441** and **442**, respectively, for memory data and Ethernet communication data input and output in the legacy virtual machine **410**.
[0129] Therefore, the present disclosure proposes a scheme for improving the system of FIG. **4**. That is, unlike FIG. **4**, virtual machines are classified into a server virtual machine and guest virtual machines for inputting and outputting various memory data and communication data not in the guest virtual machines but in the server virtual machine. This will be described with reference to FIG. **5** and subsequent figures.
[0130] FIG. **5** is a diagram showing an example of a system executed in a signal processing device according to the present disclosure.
[0131] Referring to the figure, the system **500** of FIG. **5** illustrates that the first virtual machine **520**, which is a server virtual machine, the second virtual machine **530**, which is a guest virtual machine, and the third virtual machine **540**, which is a guest virtual machine, are executed on the hypervisor **505** in the processor **175** of the signal processing device **170**.
[0132] The second virtual machine **530** may be a virtual machine for the cluster display **180***a*, and the third virtual machine **540** may be a virtual machine for the AVN display **180***b*.
[0133] That is, the second virtual machine **530** and the third virtual machine **540** may be operated for image rendering of the cluster display **180***a* and the AVN display **180***b*, respectively.
[0134] Meanwhile, the system **50** executed in the signal processing device **170** of FIG. **5** illustrates that a legacy virtual machine **510** is also executed on the hypervisor **505** in the processor **175**.
[0135] The legacy virtual machine **510** may include an interface **511** for data communication with

the memory **140** and Ethernet communication.

[0136] The figure illustrates that the interface **511** is a physical device driver; however, various modifications are possible.

[0137] Meanwhile, the legacy virtual machine **510** may further include a virtio-backend interface **512** for data communication with the second and third virtual machines **530** and **540**.

[0138] The first virtual machine **520** may include an interface **521** for input and output of audio data, radio data, USB data, and wireless communication data and an input and output server interface **522** for data communication with the guest virtual machines.

[0139] That is, the first virtual machine **520**, which is a server virtual machine, may provide inputs/outputs (I/O) difficult to virtualize with standard virtualization technology (VirtIO) to a plurality of guest virtual machines, such as the second and third virtual machines **530** and **540**.

[0140] Meanwhile, the first virtual machine **520**, which is a server virtual machine, may control radio data and audio data at a supervisor level, and may provide the data to a plurality of guest virtual machines, such as the second and third virtual machines **530** and **540**.

[0141] Meanwhile, the first virtual machine **520**, which is a server virtual machine, may process vehicle data, sensor data, and surroundings-of-vehicle information, and may provide the processed data or information to a plurality of guest virtual machines, such as the second and third virtual machines **530** and **540**.

[0142] Meanwhile, the first virtual machine **520** may provide supervisory services, such as processing of vehicle data and audio routing management.

[0143] Next, the second virtual machine **530** may include an input and output client interface **532** for data communication with the first virtual machine **520** and APIs **533** configured to control the input and output client interface **532**.

[0144] In addition, the second virtual machine **530** may include a virtio-backend interface for data communication with the legacy virtual machine **510**.

[0145] The second virtual machine **530** may receive memory data by communication with the memory **140** or Ethernet data by Ethernet communication from the virtio-backend interface **512** of the legacy virtual machine **510** through the virtio-backend interface.

[0146] Next, the third virtual machine **540** may include an input and output client interface **542** for data communication with the first virtual machine **520** and APIs **543** configured to control the input and output client interface **542**.

[0147] In addition, the third virtual machine **540** may include a virtio-backend interface for data communication with the legacy virtual machine **510**.

[0148] The third virtual machine **540** may receive memory data by communication with the memory **140** or Ethernet data by Ethernet communication from the virtio-backend interface **512** of the legacy virtual machine **510** through the virtio-backend interface.

[0149] Meanwhile, the legacy virtual machine **510** may be provided in the first virtual machine **520**, unlike FIG. **5**.

[0150] In the system **500**, CAN communication data are input and output only in the first virtual machine **520**, but may be provided to a plurality of guest virtual machines, such as the second and third virtual machines **530** and **540**, through data processing in the first virtual machine **520**. Consequently, 1:N data communication by processing of the first virtual machine **520** may be achieved.

[0151] Also, in the system **500** of FIG. **5**, audio data, radio data, USB data, and wireless communication data are input and output only in the first virtual machine **520**, but may be provided to a plurality of guest virtual machines, such as the second and third virtual machines **530** and **540**, through data processing in the first virtual machine **520**. Consequently, 1:N data communication by processing of the first virtual machine **520** may be achieved.

[0152] Also, in the system **500** of FIG. **5**, touch input to the first display **180***a* or the second display **180***b* is input only to the first virtual machine **520** and is not input to the second virtual machine

530 and the third virtual machine **540**. Information regarding the touch input is transmitted to the second virtual machine **530** or the third virtual machine **540**.

[0153] Consequently, the touch input may be rapidly and accurately processed. In addition, the touch input may be rapidly and accurately processed even though the number of virtual machines that are executed is increased.

[0154] Meanwhile, in the system **500** of FIG. **5**, the second and third virtual machines **530** and **540** may be operated based on different operating systems.

[0155] For example, the second virtual machine **530** may be operated based on a Linux OS, and the third virtual machine **540** may be operated based on a Web OS.

[0156] In the first virtual machine **520**, the shared memory **508** based on the hypervisor **505** is set for data sharing, even though the second and third virtual machines **530** and **540** are operated based on different operating systems. Even though the second and third virtual machines **530** and **540** are operated based on different operating systems, therefore, the same data or the same images may be shared in a synchronized state. Eventually, the plurality of displays **180***a* and **180***b* may display the same data or the same images in a synchronized state.

[0157] Meanwhile, the first virtual machine **520** transmits information regarding the touch input to the second virtual machine **530** or the third virtual machine **540** even though the second and third virtual machines **530** and **540** are operated based on different operating systems. Consequently, the touch input may be rapidly and accurately processed even though the second and third virtual machines **530** and **540** are operated based on different operating systems (OS).

[0158] Meanwhile, the first virtual machine **520** may include a display manager **527** configured to control overlays displayed on the first display **180***a* and the second display **180***b* through the second and third virtual machines **530** and **540**, a display layer server **529**, and a virtual overlay generator **523** configured to generate a virtual overlay.

[0159] The display layer server **529** may receive a first overlay provided by the second virtual machine **530** and a second overlay provided by the third virtual machine **540**.

[0160] Meanwhile, the display layer server **529** may transmit a virtual overlay, generated by the virtual overlay generator **523**, to at least one of the second virtual machine **530** or the third virtual machine **540**.

[0161] Meanwhile, the display manager **527** in the first virtual machine **520** may receive the first overlay provided by the second virtual machine **530** and the second overlay provided by the third virtual machine **540** through the display layer server **529**.

[0162] The display manager **527** in the first virtual machine **520** may be configured to transmit the virtual overlay, which is different from the first overlay or the second overlay, to at least one of the second virtual machine **530** or the third virtual machine **540** through the display layer server **529**.

[0163] In response thereto, the second virtual machine **530** may perform control such that the first overlay and the virtual overlay are combined and displayed on the first display **180***a*.

[0164] In addition, the third virtual machine **540** may perform control such that the second overlay and the virtual overlay are combined and displayed on the second display **180***b*.

[0165] Meanwhile, the first virtual machine **520** may include an input manager **524** configured to receive an input signal from the outside. At this time, the input signal may be an input signal from a predetermined button (start button) in the vehicle, a touch input signal, or a voice input signal.

[0166] For example, the input manager **524** in the first virtual machine **520** may receive touch input from the first display **180***a* or the second display **180***b*.

[0167] Meanwhile, the first virtual machine **520** may include a touch server **528** configured to transmit information regarding the touch input related to the touch input from the first display **180***a* or the second display **180***b* to the second virtual machine **530** or the third virtual machine **540**.

[0168] For example, in response to touch input corresponding to the first display **180***a*, the touch server **528** in the first virtual machine **520** may transmit information regarding the touch input to the second virtual machine **530**.

[0169] Meanwhile, the touch server **528** in the first virtual machine **520** may receive the touch input from the first display **180***a* or the second display **180***b*.
[0170] FIG. **6** is a diagram illustrating another example of a system executed in a signal processing device according to an embodiment of the present disclosure.
[0171] Referring to the figure, in the system **500***b* executed by the processor **175** in the signal processing device **170**, the processor **175** in the signal processing device **170** executes the first to third virtual machines **520** to **540** on the hypervisor **505** in the processor **175**, and the first virtual machine **520** in the processor **175** is configured to set he shared memory **508** based on the hypervisor **505** for transmission of data to the second and third virtual machines **530** and **540**.
[0172] For example, information regarding touch input may be illustrated as the data. Consequently, the information regarding touch input may be transmitted to the second virtual machine **530** or the third virtual machine **540**. Eventually, the touch input to the first display **180***a* or the second display **180***b* may be rapidly and accurately processed. In addition, the touch input may be rapidly and accurately processed even though the number of virtual machines that are executed is increased.
[0173] As another example, image data may be illustrated as the data. Consequently, an image may be displayed on the first display **180***a* or the second display **180***b*.
[0174] Meanwhile, in response to the same image data being shared in the shared memory **508**, the plurality of displays **180***a* and **180***b* in the vehicle may display the same data in a synchronized state.
[0175] As another example, CAN communication data, audio data, radio data, USB data, wireless communication data, or position information data may be illustrated as the data. Consequently, information regarding the data may be displayed on the first display **180***a* or the second display **180***b*.
[0176] Meanwhile, although not shown in FIG. **6**, the legacy virtual machine **510** may transmit memory data from the memory **140** or Ethernet data by Ethernet communication to the second and third virtual machines **530** and **540** using the shared memory **508** based on the hypervisor **505**. Consequently, information corresponding to the memory data or the Ethernet data may be displayed on the first display **180***a* or the second display **180***b*.
[0177] Meanwhile, the first virtual machine **520** in the system **500***b* of FIG. **6** may include a display manager **527**, a display layer server **529**, a virtual overlay generator **523**, an input manager **524**, and a touch server **528**, similarly to the first virtual machine in the system **500** of FIG. **5**.
[0178] Meanwhile, the input and output server interface **522** in the first virtual machine **520** in the system **500***b* of FIG. **6** may include a display layer server **529** and a touch server **528**, unlike FIG. **5**.
[0179] The display manager **527**, the display layer server **529**, the input manager **524**, the virtual overlay generator **523**, and the touch server **528** operate in the same manner as in FIG. **5**, such that a description thereof will be omitted.
[0180] Meanwhile, the first virtual machine **520** of FIG. **6** may further include a system manager for overall system control, a vehicle information manager for vehicle information management, an audio manager for audio control, and a radio manager for radio control.
[0181] Meanwhile, the input and output server interface **522** in the first virtual machine **520** in the system **500***b* of FIG. **6** may further include a GNSS server for GPS information input and output, a Bluetooth server for Bluetooth input and output, a Wi-Fi server for Wi-Fi input and output, and a camera server for camera data input and output.
[0182] FIG. **7** is a diagram illustrating a further example of a system executed in a signal processing device according to an embodiment of the present disclosure.
[0183] Referring to the figure, the system **500***c* executed by the processor **175** in the signal processing device of FIG. **7** is similar to the system **500***b* of FIG. **6**.
[0184] That is, like FIG. **6**, the processor **175** of FIG. **7** executes the first to third virtual machines

520 to **540** on the hypervisor **505** in the processor **175**.

[0185] In FIG. **7**, however, the display layer server **529** and the touch server **528** may be provided and executed in the first virtual machine **520** outside the input and output server interface **522**, unlike FIG. **6**.

[0186] In addition, the GNSS server for GPS information input and output, the Bluetooth server for Bluetooth input and output, the Wi-Fi server for Wi-Fi input and output, and the camera server for camera data input and output may be provided and executed in the first virtual machine **520** outside the input and output server interface **522**, unlike FIG. **6**.

[0187] That is, the display manager **527**, the display layer server **529**, the virtual overlay generator **523**, the input manager **524**, and the touch server **528** may be provided and executed in the first virtual machine **520**.

[0188] The display manager **527**, the display layer server **529**, the virtual overlay generator **523**, the input manager **524**, and the touch server **528** operate in the same manner as in FIG. **5**, such that a description thereof will be omitted.

[0189] FIGS. **8** to **9**B are diagrams referred to in the description of FIG. **5**.

[0190] First, FIG. **8** illustrates that the first to third virtual machines **520** to **540** are executed on the hypervisor **505** in the processor **175** of the system **500** according to the present disclosure and that the first virtual machine **520** in the processor **175** is configured to set he shared memory **508** based on the hypervisor **505** in order to transmit the same data to the second virtual machine **530** and the third virtual machine **540**.

[0191] Consequently, the plurality of displays **180***a* and **180***b* in the vehicle may display the same images in a synchronized state.

[0192] Meanwhile, high-speed data communication may be performed between the plurality of virtual machines. Furthermore, high-speed data communication may be performed even though the plurality of virtual machines is executed by different operating systems.

[0193] Meanwhile, the first virtual machine **520** in the processor **175** may not allocate memories corresponding in number to the virtual machines but may use a single shared memory **508**, not memory allocation in response to transmitting the data processed by the first virtual machine **520** to another virtual machine. Consequently, 1:N data communication using the shared memory **508**, not 1:1 data communication, may be performed between the virtual machines.

[0194] Meanwhile, the first virtual machine **520** in the processor **175** may include an input and output server interface **522** and a security manager **526**.

[0195] Meanwhile, the second virtual machine **530** and the third virtual machine **540** may include input and output client interfaces **532** and **542**, respectively. Consequently, high-speed data communication between the plurality of virtual machines may be performed using the input and output server interface **522** and the input and output client interfaces **532** and **542**.

[0196] The input and output server interface **522** in the first virtual machine **520** may receive requests for transmission of the same data from the input and output client interfaces **532** and **542** in the second virtual machine **530** and the third virtual machine **540**, and may transmit shared data to the shared memory **508** through the security manager **526** based thereon.

[0197] FIG. **9**A is a view illustrating transmission of shared data in more detail.

[0198] Referring to the figure, in order to transmit shared data, the input and output server interface **522** in the first virtual machine **520** transmits a request for allocation of the shared memory **508** to the security manager **526** (S**1**).

[0199] Subsequently, the security manager **526** may allocate the shared memory **508** using the hypervisor **505** (S**2**), and may write shared data in the shared memory **508**.

[0200] Meanwhile, the input and output client interfaces **532** and **542** may transmit a request for connection to the input and output server interface **522** after allocation of the shared memory **508** (S**3**).

[0201] Meanwhile, the input and output server interface **522** transmits information regarding shared memory **508** including key data to the input and output client interfaces **532** and **542** after allocation of the shared memory **508** (S**4**). At this time, the key data may be private key data for data access.

[0202] Meanwhile, the first virtual machine **520** in the processor **175** may transmit information regarding the shared memory **508** to the second virtual machine **530** and the third virtual machine **540** after setting of the shared memory **508**.

[0203] Subsequently, the input and output server interface **522** in the first virtual machine **520** is configured to generate a command or a command queue for event processing, other than data, to control distributed processing between the virtual machines (S**5**).

[0204] The figure illustrates that a command queue is generated in a command queue buffer **504** in the hypervisor **505** under control of the input and output server interface **522**. However, the present disclosure is not limited thereto, and the command queue may be generated in the first virtual machine **520**, not the hypervisor **505**, under control of the input and output server interface **522**.

[0205] Subsequently, the input and output client interfaces **532** and **542** access the command queue buffer **504** to receive the generated command queue or information regarding the command queue (S**6**).

[0206] For example, in response to the commands transmitted to the input and output client interfaces **532** and **542** being the same, the generated command queues may be the same.

[0207] As another example, in response to the commands transmitted to the input and output client interfaces **532** and **542** being different from each other, different command queues may be transmitted to the input and output client interfaces **532** and **542**.

[0208] Subsequently, the input and output client interfaces **532** and **542** may access the shared memory **508** based on the received key data (S**5**), and may copy or read the shared data from the shared memory **508** (S**7**).

[0209] Particularly, in response to the input and output client interfaces **532** and **542** receiving the same shared data, the input and output client interfaces **532** and **542** may access the shared memory **508** based on the same command queues and the same key data (S**5**), and may copy or read the shared data from the shared memory **508**.

[0210] Consequently, the second virtual machine **530** and the third virtual machine **540** may access the shared memory **508**, and may eventually share the shared data.

[0211] For example, in the case in which the shared data are image data, the second virtual machine **530** and the third virtual machine **540** may share the image data, and eventually the plurality of displays **180***a* and **180***b* in the vehicle may display the same shared images in a synchronized state.

[0212] FIG. **9**B illustrates that, by the system **500** of FIG. **9**A, the second virtual machine **530** displays image data received through the shared memory **508** on the first display **180***a*, and the third virtual machine **540** displays image data received through the shared memory **508** on the second display **180***b*.

[0213] FIG. **9**B illustrates that an image **905***a* displayed on the first display **180***a* and an image **905***b* displayed on the second display **180***b* are synchronized, whereby the same images **905***a* and **905***b* are displayed at the time of T**1**.

[0214] That is, image data processed by the first virtual machine **520** in the processor **175** are transmitted to the second virtual machine **530** and the third virtual machine **540** through the shared memory **508**, and the first image **905***a* displayed on the first display **180***a* and the second image **905***b* displayed on the second display **180***b* based on the image data may be the same. Consequently, the plurality of displays **180***a* and **180***b* in the vehicle may display the same images in a synchronized state.

[0215] FIG. **10** is an exemplary internal block diagram of a vehicle display apparatus according to an embodiment of the present disclosure, and FIGS. **11** to **14**C are diagrams referred to in the description of FIG. **10**.

[0216] First, referring to FIG. **10**, a vehicle display apparatus **100***m* according to an embodiment of

the present disclosure includes a signal processing device **170**.
[0217] The signal processing device **170** according to an embodiment of the present disclosure executes a hypervisor **505** in the processor **175**, and executes a server virtual machine **520** and a plurality of guest virtual machines **530** to **550** on the hypervisor **505**.
[0218] A first guest virtual machine **530** among the plurality of guest virtual machines **530** to **550** may operate for the first display **180***a*, a second guest virtual machine **540** may operate for the second display **180***b*, and a third guest virtual machine **550** may operate for the third display **180***c*.
[0219] The server virtual machine **520** may receive camera data from a plurality of camera devices **195** in the vehicle **200**, and may process the input camera data and transmit the processed camera data to the shared memory **508** which is set based on the hypervisor **505**.
[0220] For example, the server virtual machine **520** may perform filtering on the camera data to remove unnecessary data.
[0221] To this end, the server virtual machine **520** may execute a camera manager (not shown) and perform filtering on the camera data to remove unnecessary data.
[0222] Meanwhile, the server virtual machine **520** may further execute a monitoring manager (not shown) for driver state monitoring, an audio manager (not shown) for audio data processing, a power manger (not shown) for in-vehicle power management, and a signal manager (not shown) for vehicle signal management.
[0223] The first guest virtual machine **530** may execute an application service using the camera data. For example, the first guest virtual machine **530** may execute a navigation application.
[0224] To this end, the first guest virtual machine **530** may execute an application manager **1230** (see FIG. **12**).
[0225] Meanwhile, the first guest virtual machine **530** may further perform video or audio data processing, broadcasting data processing, voice processing, vehicle control, communication control, wireless connection with a mobile terminal, and the like.
[0226] Meanwhile, while executing an application service, the first guest virtual machine **530** may be connected to the shared memory **508** to receive the camera data stored in the shared memory **508**.
[0227] In addition, the first guest virtual machine **530** may execute a navigation application based on the camera data received from the shared memory **508**, and may control an image, corresponding to the navigation application, on the first display **180***a*.
[0228] For example, a navigation screen **1100** corresponding to the navigation application may be displayed on the first display **180** as illustrated in (a) of FIG. **11**.
[0229] The second guest virtual machine **540** may execute a first application service using the camera data. For example, the second guest virtual machine **540** may execute a location-based application such as a Point of Interest (POI).
[0230] To this end, the second guest virtual machine **540** may execute an application manager **1240** (see FIG. **12**).
[0231] Meanwhile, the second guest virtual machine **540** may further execute an on-demand application, a connected service, a translation service, and the like.
[0232] Meanwhile, while executing the first application service, the second guest virtual machine **540** may be connected to the shared memory **508** to receive the camera data stored in the shared memory **508**, may generate first data for the first application service based on the camera data, and may transmit the first data to the shared memory **508**.
[0233] In response thereto, the first guest virtual machine **530** may control an image, synthesized based on the camera data from the shared memory **508** and the first data, on the first display **180***a*. Accordingly, the navigation screen **1100** including POI information **1110** may be displayed on the first display **180***a*.
[0234] Meanwhile, while the first guest virtual machine **530** performs signal processing on the camera data for executing the navigation application, the second guest virtual machine **540** may

process data for the first application service, e.g., a location-based service, to generate first data corresponding to the location-based service data, thereby efficiently performing data processing. Particularly, data processing may be performed rapidly. Accordingly, image data processing based on camera data may be performed efficiently.
[0235] Meanwhile, the first guest virtual machine **530** may operate for the first display **180***a* based on a first operating system, and the second guest virtual machine **540** may operate for the second display **180***b* based on a second operating system different from the first operating system.
[0236] For example, the first guest virtual machine **530** may operate on the QNX operating system, and the second guest virtual machine **540** may operate on the Android operating system.
[0237] As described above, even when operating on different operating systems, the plurality of guest virtual machines **530** and **540** may share camera data using the shared memory **508** and share data generated or processed based on the shared camera data by using the shared memory **508** again, thereby efficiently performing image data processing based on the camera data. Particularly, even when operating on different operating systems, the plurality of virtual machines may efficiently process image data based on the camera data.
[0238] Meanwhile, while an image generated by synthesizing the camera data and the first data is displayed on the first display **180***a*, the second guest virtual machine **540** may control a second image, synthesized based on the camera data from the shared memory **508** and the first data, on the second display **180***b*.
[0239] Accordingly, image data processing based on camera data may be performed efficiently. Particularly, the second image synthesized in the second guest virtual machine **540** may be displayed on the second display **180***b*. That is, as illustrated in (b) of FIG. **11**, the navigation screen **1100** including the POI information **1110** may be displayed on the second display **180***b*.
[0240] Meanwhile, the first guest virtual machine **530** may transmit the first image, synthesized based on the camera data and the first data, to the shared memory **508**.
[0241] Accordingly, while the image, synthesized based on the camera data and the first data, is displayed on the first display **180***a*, the second guest virtual machine **540** may control the first image, received from the shared memory **508**, on the second display **180***b* at the same time. Accordingly, as illustrated in (b) of FIG. **11**, the navigation screen **1100** including the POI information **1110** may be displayed on the first display **180***a* and the second display **180***b* at the same time.
[0242] The third guest virtual machine **550** may execute a second application service using the camera data. For example, the third guest virtual machine **550** may execute ADAS and driver assistance application.
[0243] To this end, the third guest virtual machine **550** may execute an application manager (not shown).
[0244] Meanwhile, when executing a second application service, the third guest virtual machine **550** may be connected to the shared memory **508** to receive the camera data stored in the shared memory **508**, and may generate second data for the second application service based on the camera data and transmit the second data to the shared memory **508**.
[0245] In response thereto, the first guest virtual machine **530** may control an image, synthesized based on the camera data from the shared memory **508** and the second data, on the first display **180***a*. Accordingly, a navigation screen including ADAS information may be displayed on the first display **180***a*.
[0246] Alternatively, the first guest virtual machine **530** controls a third image, synthesized based on the camera data from the shared memory **508** and the first data and the second data, on the first display **180***a*. Accordingly, as illustrated in (c) of FIG. **11**, the navigation screen **1100** including the POI information **1110** and ADAS information **1120** may be displayed on the first display **180***a*.
[0247] Accordingly, while the first guest virtual machine **530** performs signal processing on the camera data for executing the navigation application, the second guest virtual machine **540** may

process data for an application service, e.g., a location-based service, to generate first data corresponding to the location-based service data, and the third guest virtual machine **550** may process data for a second application service, e.g., a driver assistance service, to generate second data corresponding to the driver assistance service data, thereby efficiently performing data processing. Accordingly, image data processing based on camera data may be performed efficiently.

[0248] Meanwhile, the first guest virtual machine **530** may operate for the first display **180***a* based on a first operating system, the second guest virtual machine **540** may operate for the second display **180***b* based on a second operating system different from the first operating system, and the third guest virtual machine **550** may operate for the third display **180***c* based on a third operating system different from the first operating system and the second operating system.

[0249] For example, the first guest virtual machine **530** may operate on the QNX operating system, the second guest virtual machine **540** may operate on the Android operating system, and the third guest virtual machine **550** may operate on the Linux operating system.

[0250] As described above, even when operating on different operating systems, the plurality of guest virtual machines **530**, **540**, and **550** may share camera data using the shared memory **508** and share data generated or processed based on the shared camera data by using the shared memory **508** again, thereby efficiently performing image data processing based on the camera data. Particularly, even when operating on different operating systems, the plurality of virtual machines may efficiently perform image data processing based on the camera data.

[0251] Meanwhile, while the image, synthesized based on the camera data and the first data, is displayed on the first display **180***a*, the second guest virtual machine **540** may control the second image, synthesized based on the camera data from the shared memory **508** and the first data, on the second display **180***b*, and the third guest virtual machine **550** may control a fourth image, synthesized based on the camera data from the shared memory **508** and the second data, on the third display **180***c*.

[0252] That is, as illustrated in (b) of FIG. **11**, while the navigation screen **1100** including the POI information **1110** is displayed on the first display **180***a* and the second display **180***b*, the navigation screen **1100** including the ADAS information **1120** may be displayed on the third display **180***c*. Accordingly, the displays may display images corresponding to the respective displays while efficiently performing image data processing based on the camera data.

[0253] Meanwhile, while the image, synthesized based on the camera data and the first data, is displayed on the first display **180***a*, the second guest virtual machine **540** may control a fifth image, synthesized based on the camera data from the shared memory **508** and the first data and the second data, on the second display **180***b*, and the third guest virtual machine **550** may control a sixth image, synthesized based on the camera data from the shared memory **508** and the first data and the second data, on the third display **180***c*.

[0254] That is, as illustrated in (c) of FIG. **11**, the navigation screen **1100** including the POI information **1110** and the ADAS information **1120** may be displayed on the first display **180***a*, the second display **180***b*, and the third display **180***c*. Accordingly, the respective displays **180***a*, **180***b*, and **180***c* may display the same image while efficiently performing image data processing based on the camera data.

[0255] Meanwhile, the first guest virtual machine **530** may transmit the synthesized third image to the shared memory **508**, and at least one of the second guest virtual machine **540** or the third guest virtual machine **550** may be configured to display the synthesized third image from the shared memory **508** on at least one of the second display **180***b* or the third display **180***c*.

[0256] That is, as illustrated in (c) of FIG. **11**, the navigation screen **1100** including the POI information **1110** and the ADAS information **1120** may be displayed on the first display **180***a*, the second display **180***b*, and the third display **180***c*. Accordingly, the respective displays **180***a*, **180***b*, and **180***c* may display the same image while efficiently performing image data processing based on

the camera data.
[0257] FIG. **11** is a diagram illustrating various examples of a navigation screen.
[0258] Referring to FIG. **11**, (a) of FIG. **11** illustrates a signal-processed navigation screen **1100** in response to executing a navigation application in the first guest virtual machine **530**.
[0259] Meanwhile, if the second guest virtual machine **540** executes the first application for displaying POI information, the navigation screen **1100** including the POI information **1110** may be displayed on the first display **180***a*, as illustrated in (b) of FIG. **11**.
[0260] In this case, a POI item **1101** may be highlighted among a plurality of items in the navigation screen **1100**.
[0261] Meanwhile, if the third guest virtual machine **550** executes the second application for displaying ADAS information, the navigation screen **1100** including the POI information **1110** and the ADAS information **1120** may be displayed on the first display **180***a*, as illustrated in (c) of FIG. **11**.
[0262] In this case, the POI item **1101** and an ADAS item **1103** may be highlighted among a plurality of items in the navigation screen **1100**.
[0263] Meanwhile, if the second guest virtual machine **540** further executes a third application for displaying translation information, the second guest virtual machine **540** may generate third data for the third application service based on the camera data and transmit the third data to the shared memory **508**.
[0264] In response thereto, the first guest virtual machine may control the navigation screen **1100**, including the POI information **1110**, the ADAS information **1120**, and translation information **1130**, on the first display **180***a*.
[0265] That is, as illustrated in (d) of FIG. **11**, the navigation screen **1100** including the POI information **1110**, the ADAS information **1120**, and the translation information **1130** may be displayed on the first display **180***a*.
[0266] In this case, the POI item **1101**, the ADAS item **1103**, and the translation item **1105** may be highlighted among a plurality of items in the navigation screen **1100**.
[0267] Meanwhile, unlike FIG. **11**, it is also possible that the translation information **1130** is processed by the first guest virtual machine **530**, and the POI information **1110** and the ADAS information **1120** are processed by the second guest virtual machine, which will be described below with reference to FIG. **12**.
[0268] FIG. **12** is an exemplary internal block diagram of a signal processing device according to an embodiment of the present disclosure.
[0269] Referring to FIG. **12**, a signal processing device **170***ma* according to an embodiment of the present disclosure may include a processor **175** configured to execute the server virtual machine **520** and the plurality of guest virtual machines **530** and **540**, and shared memories **508***a* and **508***b*.
[0270] The first guest virtual machine **530** among the plurality of guest virtual machines **530** and **540** may execute each of a navigation application (not shown) and a translation application **1232**. The translation application **1232** may include a data sharing control library **1233**.
[0271] The first guest virtual machine **530** may execute an application manager **1230** for executing the navigation application (not shown) and the translation application **1232**.
[0272] The second guest virtual machine **540** among the plurality of guest virtual machines **530** and **540** may execute an ADAS application **1242**. The ADAS application **1242** may include a data sharing control library **1243**.
[0273] The second guest virtual machine **540** may execute the application manager **1240** for executing the ADAS application **1242**.
[0274] Meanwhile, the application managers **1230** and **1240** in the first guest virtual machine **530** or the second guest virtual machine **540** may transmit an event request to the server virtual machine **520**, and the server virtual machine **520** may control sharable camera data among the processed camera data to be transmitted to the shared memory **508** based on the even request.

[0275] Meanwhile, the server virtual machine **520** may process the camera data.

[0276] For example, the server virtual machine **520** may perform processing operations, such as object detection, picture stabilization, and the like.

[0277] Meanwhile, the server virtual machine **520** may execute a data sharing manager **1210** for sharing camera data, and a camera processing manger **1220** for processing camera data.

[0278] The data sharing manager **1210** may execute or include a Camera Operation Category table **1212** and a Camera processing data application association table **1214**.

[0279] The Camera Operation Category table **1212** may be a table in which camera data processing operations used in a camera application are classified into operations and stored.

[0280] By using this table **1212**, commonly used camera data processing may be confirmed, and camera processing pipeline may be reconfigured based on the confirmation result.

[0281] The Camera processing data application association table **1214** may be a table for storing a state in which camera data is shared between applications or the camera data is commonly used.

[0282] The camera processing manager **1220** may execute a Camera processing pipeline information table **1222**, a Camera processing pipeline **1224**, and a Camera Processing Data Shared Cache **1226**.

[0283] The Camera processing pipeline information table **1222** may be a table including information required for actual processing for each camera processing operation. By using this table, processing suitable for each stage of the pipeline may be performed.

[0284] The Camera processing pipeline **1224** operates for transmitting camera data.

[0285] When the processed camera data is shared between applications, the Camera Processing Data Shared Cache **1226** puts the processed data and notifies an application that has requested the data.

[0286] FIGS. **13**A to **13**E are diagrams referred to in the description of operation of FIG. **12**.

[0287] For example, if the translation application **1232** is executed in the first guest virtual machine **530**, the application manager **530** of the first guest virtual machine **530** transmits an event notification FLa to the server virtual machine **520**.

[0288] Then, the data sharing manager **1210** in the server virtual machine **520** checks whether camera data is sharable in the translation application **1232**, and if camera data is sharable, the data sharing manager **1210** may transmit sharable information FLb to the camera processing manager **1220** as illustrated in FIG. **13**B.

[0289] The Camera processing pipeline **1224** in the camera processing manager **1220** may write processed camera data FLc to the shared memory **508***a* based on the sharable information FLb, as illustrated in FIG. **13**C.

[0290] The shared memory **508***a* may include a camera data frame Dta and camera processing data Dtb.

[0291] Meanwhile, the camera data FLc written to the shared memory **508***a* may be transmitted to the data sharing control library **1233** in the translation application **1232**. Accordingly, camera data may be simply shared.

[0292] Meanwhile, the translation application **1232** in the first guest virtual machine **530** may be configured to perform data processing based on the received camera data, and to display the translation information **1130** in (d) of FIG. **11**.

[0293] Then, if the ADAS application **1242** is executed in the second guest virtual machine **540**, the application manager **530** of the first guest virtual machine **530** transmits a second event notification FLd to the server virtual machine **520**.

[0294] Then, the data sharing manager **1210** in the server virtual machine **520** checks whether camera data is sharable in the ADAS application **1242**, and if camera data is sharable, the data sharing manager **1210** may transmit sharable information to the camera processing manager **1220**.

[0295] The Camera processing pipeline **1224** and the Camera Processing Data Shared Cache **1226** in the camera processing manager **1220** may write processed camera data FLe and FLf to the

shared memory **508***b* based on the sharable information, as illustrated in FIG. **13**E.
[0296] The shared memory **508***b* may include a camera data frame Dtac and camera processing data Dtd.
[0297] Meanwhile, the camera data FLe and FLf written to the shared memory **508***a* may be transmitted to the data sharing control library **1243** in the ADAS application **1242**. Accordingly, camera data may be simply shared.
[0298] Meanwhile, the ADAS application **1242** in the second guest virtual machine **540** may be configured to perform data processing based on the received camera data, and to display the ADAS information **1120** in (c) of FIG. **11**.
[0299] FIGS. **14**A and **14**B are diagrams illustrating various examples of a list of camera operation categories.
[0300] FIG. **14**A is a diagram illustrating a list of applications related to object detection based on camera data.
[0301] Referring to FIG. **14**A, the server virtual machine **520** may detect objects, such as roads, road lines, traffic signs, pedestrians, and the like.
[0302] The detected objects may be used in AR HUD applications, POI applications, Traffic Sign Recognition (TSR) applications, translation applications, ADAS applications, and the like.
[0303] FIG. **14**B is a diagram illustrating a list of applications related to Picture Stabilization based on camera data.
[0304] Referring to FIG. **14**B, the server virtual machine **520** may perform picture stabilization, and camera data on which picture stabilization is performed may be used for all image-related applications.
[0305] FIGS. **15**A to **15**C are an exemplary internal block diagram of the signal processing device **170** for explaining operation of FIG. **14**A or FIG. **14**B.
[0306] First, referring to FIG. **15**A, the signal processing device **170** may execute or include an object detection processor **1515** for object detection from camera data, and a picture stabilization processor **1510** for picture stabilization from the camera data.
[0307] For example, the object detection processor **1515** and the picture stabilization processor **1510** may be executed in the server virtual machine **520**.
[0308] That is, the server virtual machine **520** may perform object detection and picture stabilization by executing the object detection processor **1515** and the picture stabilization processor **1510**, respectively, based on the camera data.
[0309] The first guest virtual machine **530** in the signal processing device **170** may execute a navigation application **1525**, and may receive camera data, on which the picture stabilization is performed, via the shared memory **508**.
[0310] Then, the first guest virtual machine **530** in the signal processing device **170** may execute the data processor **1520** to perform signal processing on the camera data on which the picture stabilization is performed, and may transmit the processed camera data to the navigation application **1525**.
[0311] FIG. **15**B is a diagram illustrating an example in which a TSR application and an ADAS application are executed in the signal processing device **170**.
[0312] Referring to FIG. **15**B, the second guest virtual machine **540** in the signal processing device **170** may execute a TSR application **1532** and an ADAS application **1534**, and may receive a camera data, having a detected object, via the shared memory **508**.
[0313] Then, the second guest virtual machine **540** in the signal processing device **170** may execute the data processor **1530** to perform signal processing on the camera data having the detected object, and may transmit the processed camera data, having the detected object, to the TSR application **1532** and the ADAS application **534**.
[0314] FIG. **15**C is a diagram illustrating an example in which a POI application is executed in the signal processing device **170**.

[0315] Referring to FIG. **15**C, the third guest virtual machine **540** in the signal processing device **170** executes a POI application **1539**, and may receive camera data, in which an object is detected, via the shared memory **508**.
[0316] Then, the third guest virtual machine **540** in the signal processing device **170** may execute the data processor **1538** to perform signal processing on the camera data in which an object is detected, and may transmit the processed camera data, having the detected object, to the POI application **1539**.
[0317] Meanwhile, the camera data having the detected object and processed by the data processor **1538** may also be transmitted to the Camera Processing Data Shared Cache **1542** in the server virtual machine **520**.
[0318] Referring to FIGS. **15**A to **15**C, the signal processing device **170** according to an embodiment of the present disclosure may perform common processing for applications using the same camera data, and then may perform separate processing operations for the respective applications.
[0319] Meanwhile, the signal processing device **170** according to an embodiment of the present disclosure may also share processing results if the same camera data is shared between other applications.
[0320] Meanwhile, the signal processing device **170** according to an embodiment of the present disclosure may share the camera data by using the Camera Processing Data Shared Cache **1542** and the shared memory **508** in response to a request for camera data used in the second application.
[0321] It will be apparent that, although the preferred embodiments have been shown and described above, the present disclosure is not limited to the above-described specific embodiments, and various modifications and variations can be made by those skilled in the art without departing from the gist of the appended claims. Thus, it is intended that the modifications and variations should not be understood independently of the technical spirit or prospect of the present disclosure.

Claims

1. A signal processing device comprising a processor configured to perform signal processing for a display located in a vehicle, wherein the processor is configured to execute a plurality of virtual machines on a hypervisor in the processor, wherein the first virtual machine among the plurality of virtual machines is configured to, in response to receiving a request to share camera data from a first application executed in a second virtual machine, write camera data processed in camera processing data shared cache to a shared memory, and notify the first application.
2. The signal processing device of claim 1, wherein the first virtual machine is configured to, in response to receiving a request to share camera data from a second application executed in a third virtual machine, write camera data processed in camera processing data shared cache to a shared memory, and notify the second application.
3. The signal processing device of claim 1, wherein the first virtual machine is configured to perform picture stabilization based on input camera data, perform object detection based on a first camera data on which the picture stabilization is performed, and transmit the first camera data and a second camera data on which the object detection is performed to the shared memory, wherein the second virtual machine is configured to receive the first camera data from the shared memory and display an image corresponding to the first camera data.
4. The signal processing device of claim 3, wherein a third virtual machine among the plurality of virtual machines is configured to receive the second camera data from the shared memory, to generate first data for the first application service based on the second camera data, and to transmit the first data to the shared memory; and wherein the second virtual machine is configured to output an image synthesized based on the first camera data and the first data from the shared memory.
5. The signal processing device of claim 1, wherein the first virtual machine is configured to

operate based on a first operating system, and the second t virtual machine is configured to operate display based on a second operating system different from the first operating system.
6. The signal processing device of claim 4, wherein the third virtual machine is configured to output a second image synthesized based on the second camera data and the first data from the shared memory.
7. The signal processing device of claim 4, wherein the first virtual machine is configured to operate based on a first operating system, and the second virtual machine is configured to operate based on a second operating system different from the first operating system, and the third virtual machine is configured to operate based on a third operating system different from the first operating system and the second operating system.
8. The signal processing device of claim 1, wherein an application manager in the second virtual machine is configured to transmit an event request to the first virtual machine, wherein the first virtual machine is configured to transmit sharable camera data among the processed camera data to the shared memory based on the event request.
9. The signal processing device of claim 1, wherein the first virtual machine is configured to execute a data sharing manager for sharing the data and a camera processing manager for processing the camera data.
10. The signal processing device of claim 9, wherein the camera processing manager is configured to execute camera processing pipeline information table, camera processing pipeline, and the camera processing data shared cache.
11. The signal processing device of claim 10, wherein the camera processing pipeline information table corresponds to a table including information required for actual processing for each camera processing operation.
12. A signal processing device comprising a processor configured to perform signal processing for at least one display located in a vehicle, wherein the processor is configured to execute a first virtual machine and a second virtual machine that operates on different operating systems of the first virtual machine on a hypervisor in the processor, wherein the first virtual machine is configured to, in response to receiving a request to share camera data from a first application executed in a second virtual machine, share data processed in camera processing data shared cache, and notify the first application.
13. The signal processing device of claim 12, wherein the shared data comprises at least one of camera data, radar data, lidar data, audio data, or image data.
14. A vehicle display apparatus comprising: a plurality of displays; and a signal processing device comprising a processor configured to perform signal processing for the plurality of displays, wherein the signal processing device comprises a processor configured to perform signal processing for a plurality of displays located in a vehicle, wherein the processor is configured to execute a plurality of virtual machines on a hypervisor in the processor, wherein a first virtual machine among the plurality of virtual machines is configured to, in response to receiving a request to share camera data from a first application executed in a second virtual machine, write camera data processed in camera processing data shared cache to a shared memory, and notify the first application, wherein the second virtual machine and a third virtual machine among the plurality of guest virtual machines are configured to operate for a first display and a second display, respectively.
15. The vehicle display apparatus of claim 14, wherein the first virtual machine is configured to, in response to receiving a request to share camera data from a second application executed in the third virtual machine, write camera data processed in camera processing data shared cache to a shared memory, and notify the second application.
16. The vehicle display apparatus of claim 15, wherein the second virtual machine is configured to operate for the first display based on a first operating system, and the second guest virtual machine is configured to operate for the second display based on a second operating system different from

the first operating system.

17. The vehicle display apparatus of claim 15, wherein the second guest virtual machine is configured to display a second image, synthesized based on the second camera data and the first data from the shared memory, on the second display.